

Integrating Security into the IaC Pipeline

Preparing files for GitHub Repository

Template file to define what is needed.

```
! buildspec.yml M    ! template.yaml X
room8CICD > ! template.yaml
cadygal, 1 hour ago | 1 author (cadygal)
1  AWSTemplateFormatVersion: '2010-09-09'
2  Description: Basic S3 Bucket Template for CI/CD + Conformity Scanner
3
4  Resources:
5    SampleS3Bucket:
6      Type: AWS::S3::Bucket
7      Properties:
8        BucketName: !Sub 'conformity-scanner-bucket-${AWS::AccountId}-${AWS::Region}'
9        VersioningConfiguration:
10         Status: Enabled
11
12  Outputs:
13    BucketName:
14      Description: Name of the created S3 bucket
15      Value: !Ref SampleS3Bucket
16
17  BucketEncryption:
18    ServersideEncryptionConfiguration:
19      - ServersideEncryptionByDefault:
20        SSEAlgorithm: AES256
21
```

Creating API keys on Conformity to prepare our Builds spec.yml file.

Trend Cloud One™ > Administration > API Keys

Account Settings
Users
Roles
API Keys
Identity Providers
Audit Log

API Keys

New Delete

API Key Alias	Role	Enabled	Created	Last Modified	Last Used
room8	Full Access	✓	May 30, 2025, 20:59	May 30, 2025, 20:59	

API Key Properties

API Key Alias: room8

API Key ID: tmc12xpd4fFaXvGYvRLQA8rMl2pPoI

Description:

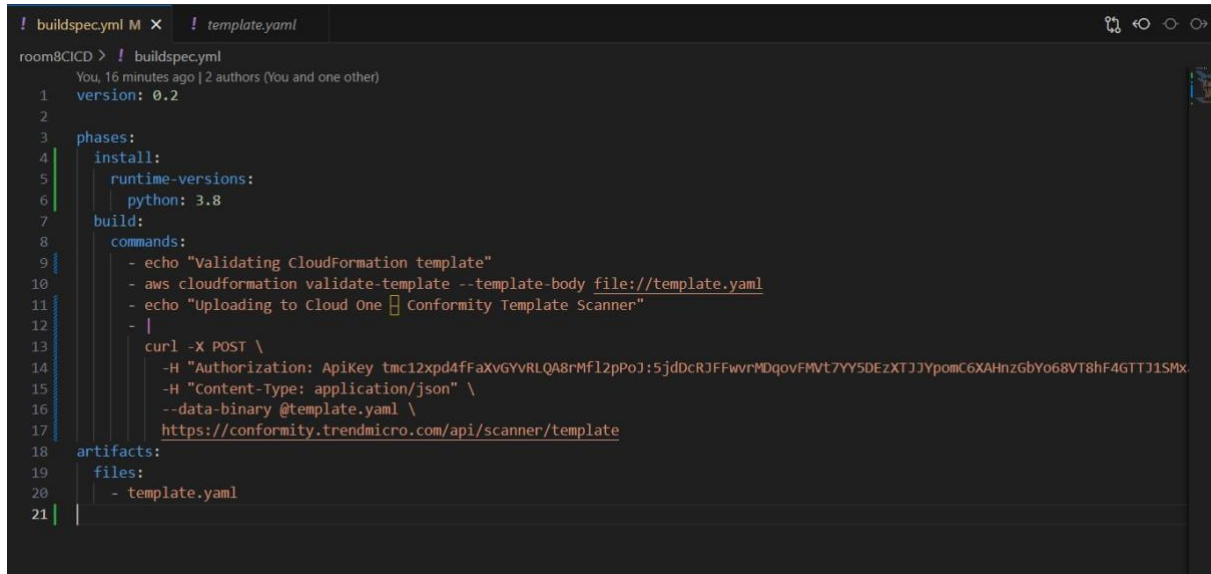
Created: May 30, 2025, 20:59

Last Modified: May 30, 2025, 20:59

Last Used:

Save

Buildspec file for CloudFormation Template



```
! buildspec.yml M x ! template.yaml
room8CICD > ! buildspec.yml
You, 16 minutes ago | 2 authors (You and one other)
1 version: 0.2
2
3 phases:
4   install:
5     runtime-versions:
6       python: 3.8
7     build:
8       commands:
9         - echo "Validating CloudFormation template"
10        - aws cloudformation validate-template --template-body file://template.yaml
11        - echo "Uploading to Cloud One [icon] Conformity Template Scanner"
12        - |
13          curl -X POST \
14            -H "Authorization: ApiKey tmc12xpd4fFaXvGYVRLQA8rMf12pPoJ:5jdDcR3JFFwvrMDqovFMvt7YY5DEzXTJJYpomC6XAHnzGbYo68VT8hF4GTTJ1SMx..."
15            -H "Content-Type: application/json" \
16            --data-binary @template.yaml \
17            https://conformity.trendmicro.com/api/scanner/template
18 artifacts:
19   files:
20     - template.yaml
21
```

Pushing files from VSCode to Github Repository

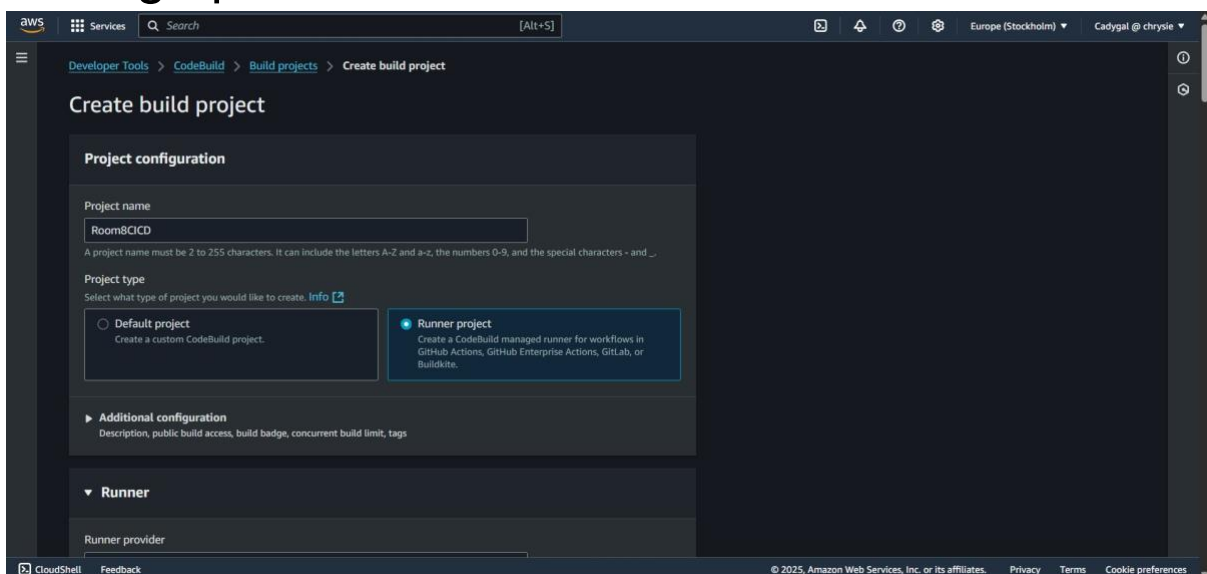
```

PS C:\Users\chrys\Desktop\room8CICD> git add .
>> git commit -m "Initial CloudFormation template"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\chrys\Desktop\room8CICD> git push -u origin main
info: please complete authentication in your browser...
To https://github.com/cadygal/room8CICD.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/cadygal/room8CICD.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
PS C:\Users\chrys\Desktop\room8CICD> git pull
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 2.11 KiB | 143.00 KiB/s, done.
From https://github.com/cadygal/room8CICD
 7f82017..5759ca9  main      -> origin/main
Updating 7f82017..5759ca9
Fast-forward
 template.yaml | 20 ++++++++-----
 1 file changed, 18 insertions(+), 2 deletions(-)
PS C:\Users\chrys\Desktop\room8CICD> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date

```

Setting up Code build to run our code



aws

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Europe (Stockholm)

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Runner

Runner provider

GitHub

Credential

✔ Your account is successfully connected by using an AWS managed GitHub App. Manage account credentials.

☐ Use override credentials for this project only

Runner location

☒ Repository
Trigger runner builds on repository webhook events

☐ Organization
Trigger runner builds on organization webhook events

☐ Enterprise
Trigger runner builds on enterprise webhook events

Repository

Q https://github.com/cadygal/room8CICD

X

↺

The repository URL of the GitHub Actions repository.

Runner Configuration Sample

The following GitHub Actions workflow YAML can be used to trigger jobs on the PUSH event for this CodeBuild project. For more information about CodeBuild GitHub Actions label syntax, visit [CodeBuild-Hosted GitHub Actions Supported Label Overrides](#).

```
name: Hello World
on: [push]
jobs:
  Hello-World-Job:
```

📄

CloudShell

Feedback

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Environment image

☒ Managed image
Use an image managed by AWS CodeBuild

☐ Custom image
Specify a Docker image

Running mode

☒ Container
Running on Docker container

☐ Instance
Running on EC2 instance directly

Operating system

Ubuntu

Runtime(s)

Standard

Image

aws/codebuild/standard:7.0

Image version

Always use the latest image for this runtime version

Service role

☒ New service role
Create a service role in your account

☐ Existing service role
Choose an existing service role from your account

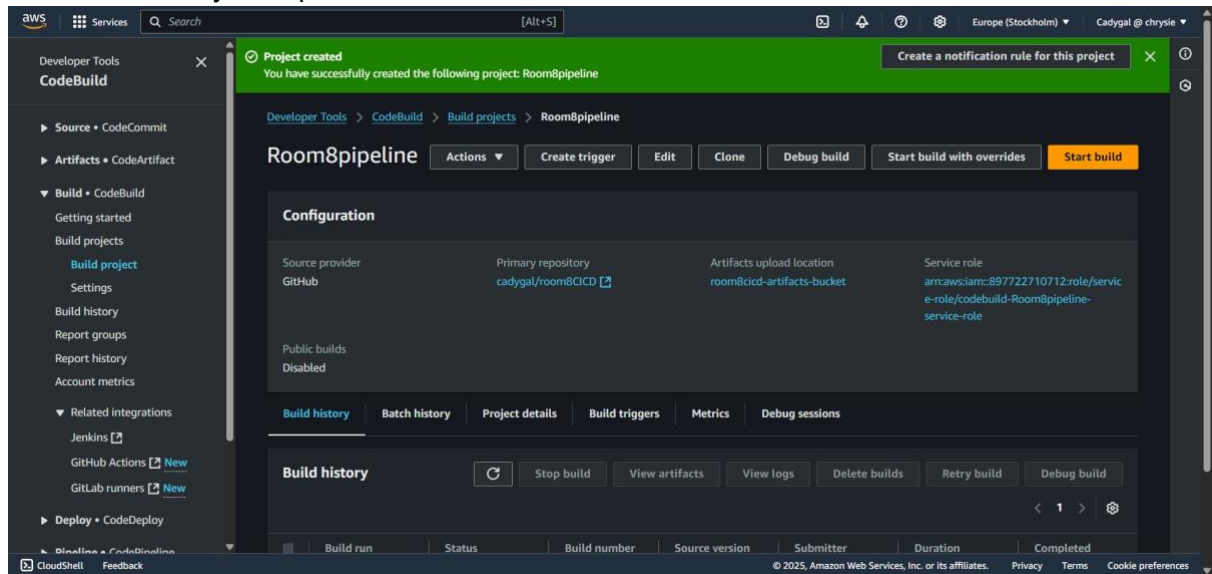
Role name

CloudShell

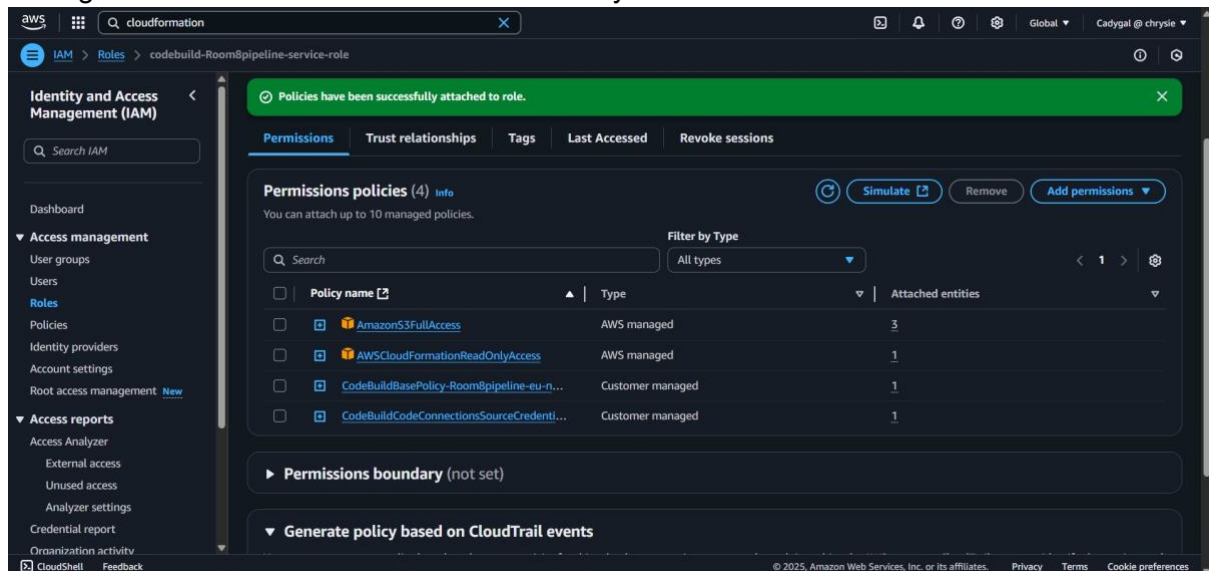
Feedback

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Build Successfully set up



Adding S3 and CloudFormation access to newly created role for CodeBuild



Running our template through the conformity template scanner to detect errors.

Trend Cloud One™ > Conformity

AccountsAdministrationProfilesTemplate Scanner

Chrysolite Yebsoah

Template Scanner allows you to run Cloud Posture Rules on your Infrastructure-as-Code template. This enables you to add preventative security and governance controls to your workflow, allowing your team to identify and remediate issues prior to any services and resources being launched.

Template Scanner is also supported via our API. For more information, read about Template Scanner in our Help Pages.

CloudFormationTerraform

CloudFormation Template

template.yaml (469 bytes)

Browse

Default rule settings

Profile rule settings

Account rule settings

Upload and scan

Template Scan results

NOTE: Resolution steps provided are for workflows via CLI or Console, however, you can use the steps as a guide for resolution within CloudFormation.

Rule	Service	Categories	Risk Level	Counts
+ S3 Bucket Public 'READ' Access	S3	Security	Very High	SUCCESS 1
+ S3 Bucket Public 'READ_ACP' Access	S3	Security	Very High	SUCCESS 1
+ S3 Bucket Public 'WRITE' ACL Access	S3	Security	Very High	SUCCESS 1
+ S3 Bucket Public 'WRITE_ACP' Access	S3	Security	Very High	SUCCESS 1
+ S3 Bucket Public 'FULL_CONTROL' Access	S3	Security	Very High	SUCCESS 1
+ S3 Bucket Authenticated Users 'READ' Access	S3	Security	Very High	SUCCESS 1

Trend Cloud One™ > Conformity

AccountsAdministrationProfilesTemplate Scanner

Chrysolite Yebsoah

+ S3 Bucket Authenticated Users 'READ_ACP' Access	S3	Security	Very High	SUCCESS 1	
+ S3 Bucket Authenticated Users 'WRITE' Access	S3	Security	Very High	SUCCESS 1	
+ S3 Bucket Authenticated Users 'WRITE_ACP' Access	S3	Security	Very High	SUCCESS 1	
+ S3 Bucket Authenticated Users 'FULL_CONTROL' Access	S3	Security	Very High	SUCCESS 1	
+ S3 Bucket Logging Enabled	S3	Security	Medium	FAILURE 1	Resolve...
+ S3 Bucket Versioning Enabled	S3	Reliability	Low	SUCCESS 1	
+ S3 Bucket MFA Delete Enabled	S3	Security	Low	FAILURE 1	Resolve...
+ Server Side Encryption	S3	Security	High	SUCCESS 1	
+ Secure Transport	S3	Security	Medium	FAILURE 1	Resolve...
+ DNS Compliant S3 Bucket Names	S3	Performance Efficiency	Low	SUCCESS 1	
+ S3 Buckets Lifecycle Configuration	S3	Security, Cost Optimization, Sustainability	Low	FAILURE 1	Resolve...
+ S3 Object Lock	S3	Security	Low	FAILURE 1	Resolve...
+ S3 Transfer Acceleration	S3	Performance Efficiency	Low	FAILURE 1	Resolve...
+ Enable S3 Block Public Access for S3 Buckets	S3	Security	Medium	FAILURE 1	Resolve...
+ Tags	Resource Group	Security, Reliability, Performance Efficiency, Cost Optimization, Operational Excellence, Sustainability	Low	FAILURE 1	Resolve...

Creating a Code Pipeline

The screenshot shows the AWS CodePipeline console with the following configuration fields filled out:

- CodeConnections ARN for your source repository:** `arn:aws:codeconnections:eu-north-1:897722710712:connection/4c13597b-1948-4fa-`
- FullRepositoryId:** `cadygal/room8CICD`
- BranchName:** `main`
- CodePipelineName:** `DeployToCloudFormationService`
- StackName:** `Room8Stack`
- TemplatePath:** `template.yaml`
- OutputFileName:** `output.json`

The bottom of the console shows the **CloudFormationResourcePermissions** section, which is currently empty.

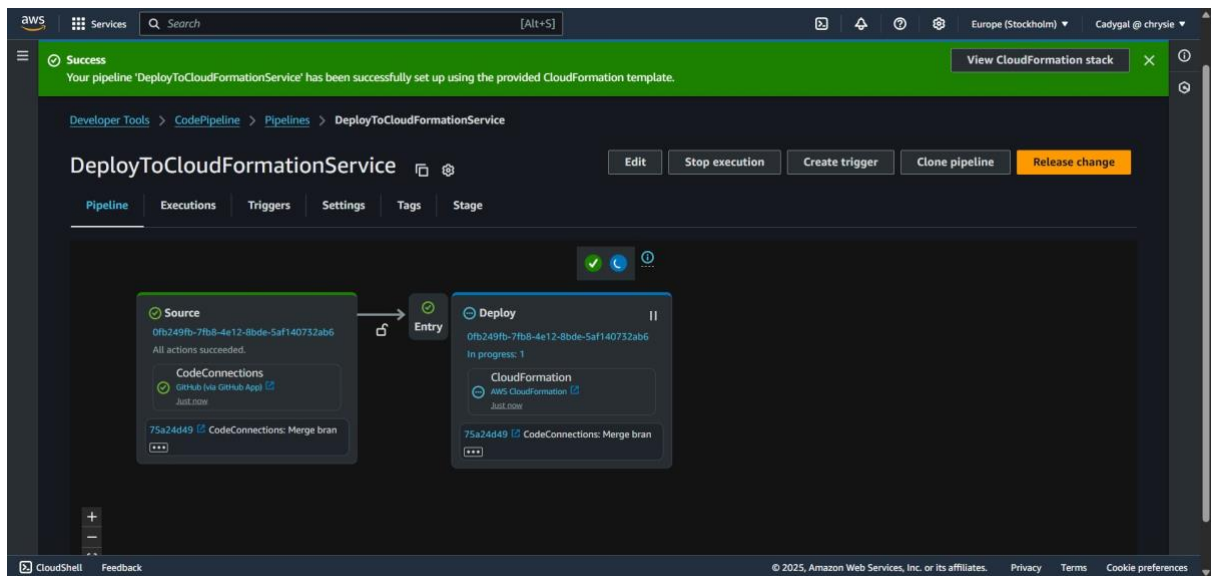
The screenshot shows the AWS CodePipeline console with the following configuration fields filled out:

- OutputFileName:** `output.json`
- CloudFormationResourcePermissions:** A JSON policy is provided:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "PermissionsForCloudFormation",
6       "Effect": "Allow",
7       "Action": "*",
8       "Resource": "*"
9     }
10  ]
11 }
12
```
- RetentionPolicy:** `Delete`
- Advanced settings:** `TemplateConfiguration, ParameterOverrides`
- TemplateConfiguration - optional:** (Empty field)
- ParameterOverrides - optional:** (Empty field)

The bottom of the console shows the **CloudFormationResourcePermissions** section, which is currently empty.

Code pipeline successfully deployed.



Group Members

1. Richmond Kobby Nyarko
2. Chrysolite Yeboah